Soudeep Deb

Associate Professor and Chairperson
Decision Sciences Area
Indian Institute of Management Bangalore, India.

Research group: signal-lab-by-deb.in
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Current position

Research Interests Time series, Spatio-temporal models, Nonparametric methods, Statistical machine learning, Sports analytics, Applications of statistics in real estate, finance, environmental science, and other fields.

Experience

Indian Institute of Management Bangalore, Bengaluru, KA, India. Mar 2020 - Present

- Chairperson, Decision Sciences Area. Apr 2024 - Present
- Associate Professor, Decision Sciences Area. Jul 2024 - Present
- Assistant Professor, Decision Sciences Area. Mar 2020 - Jul 2024

NBC Universal Media, LLC., New York, NY, USA. Sep 2018 - Feb 2020

- Senior Lead Data Scientist, Decision Sciences Division.

Education

University of Chicago, Chicago, IL, USA.

Ph.D., Statistics Aug 2018

- Thesis: Irregular spaced random field, Spatio-temporal data and Clustering of time series
- Advisory committee: Dr. Wei Biao Wu, Dr. Ruey S. Tsay, Dr. Michael L. Stein

Indian Statistical Institute, Kolkata, WB, India.

Master of Statistics (M. Stat.) May 2013

- First Division with Distinction, Specialization: Mathematical Statistics and Probability
- Thesis: Association analysis for identifying rare genetic variants, Advisor: Dr. Saurabh Ghosh

Bachelor of Statistics (B. Stat.) May 2011

- First Division with Distinction

Books

- (With S. Das) Business Analytics: Data to Decisions. First edition in 2025, published by Universities Press (India) Private Limited. ISBN: 9789393330994. [Amazon link]
- (With D. Dey) Mathematical Techniques for Competitive Examinations. First edition in 2023, published by Orient Blackswan Pvt Ltd. ISBN: 9393330107. [Amazon link]

Research grants

India-Israel Joint Research Cooperation (IIJRC) Grant, INR 1.6 million. 2024 – 2026

- From Department of Science & Technology, India, and Ministry of Innovation, Science & Technology, Israel.
- Co-PI: Prof. Itai Dattner, University of Haifa, Israel.
- Project: Hybrid-AI Quantile Regression Combining Nonparametric Statistical Methods and Physics-Informed Neural Networks for Analysing Spatio-temporal Climate Data.

Research Challenge Grant from IIM Bangalore, India, INR 1.5 million. 2025 – 2026

- Co-PI: Prof. Anand Deo, IIM Bangalore, India.
- Project: Estimating spatial extremes using EVT-based scaling.

Research Travel Grant from London Mathematical Society, UK, GBP 2200. 2024

- Offered under Scheme 5 'Collaborations with Developing Countries' Grant.

Research Seed Grant from IIM Bangalore, India, INR 0.5 million. 2024 – 2025

- Project: Nonparametric methods of structural break detection in time series.

Research Seed Grant from IIM Bangalore, India, INR 0.3 million. 2021 – 2023

- Project: New techniques to analyze categorical and discrete time series data.

Young Faculty Research Chair Grant from IIM Bangalore, India, INR 0.9 million. 2020 – 2023

Students

PhD thesis advisor/co-advisor:

- Anchal Soni, Decision Sciences, Indian Institute of Management Bangalore. Feb 2023
  Thesis: Methods of analyzing structural breaks in multivariate time series: Applications to financial data.
- Siddharth Rawat, Decision Sciences, Indian Institute of Management Bangalore. May 2023
  Thesis: Spatio-temporal models in epidemiology and climate change.
- Kapil Gupta, Decision Sciences, Indian Institute of Management Bangalore. Mar 2025
  Thesis: Analyzing house price dynamics using novel spatio-temporal methods.
- Archi Roy, Dept of Mathematics, Indian Institute of Science Education & Research Pune. Aug 2025
  Thesis: Nonparametric methods for short and long-range dependent time series: Applications in financial and biomedical datasets.
- Chinmay Divekar, Decision Sciences, Indian Institute of Management Bangalore. Sep 2026
  Thesis: Novel methods for within-game analytics problems in sports.
- Kunal Rai, Decision Sciences, Indian Institute of Management Bangalore. Nov 2026
  Thesis: Nonparametric Quantile Regression in Time Series: Theory and Applications.
- Amrutha Seshagiri, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2028
  Thesis: Essays on tourism analytics.
- Lizan Meryl Pereira, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2029
  Thesis: Methods in Spatial Statistical Modelling for Environmental Risk.

Doctoral committee member:

- M Mareeswaran, Finance & Accounting, Indian Institute of Management Bangalore. Mar 2025
- Sajad S Santhosh, Public Policy, Indian Institute of Management Bangalore. Jul 2025
- Sabhya Rai, Economics, Indian Institute of Management Bangalore. Mar 2026
- Kalaga Venkata Laxmana Tejaswi, Marketing, Indian Institute of Management Bangalore. (Exp) 2028
- Mantu Prasad Gupta, Decision Sciences, Indian Institute of Management Bangalore. (Exp) 2028

MS thesis advisor:

- Snehil Maurya, Indian Institute of Science Education & Research Mohali. 2026
- Ashish Kumar Yadav, Indian Institute of Science Education & Research Mohali. (Exp) 2027

Papers under review

- Banerjee, S., **Deb, S.** (2026). Projection diagnostics for directional asymmetry and tail-ratio departure in multivariate data. [preprint]
- Roy, S., **Deb, S.**, Karmakar, S., Roy, R. (2026). Nonparametric regression of spatio-temporal data using infinite-dimensional covariates. [preprint]
- Basu, T., **Deb, S.**, Roy, R. (2026). Optimising football transfer strategy under budget constraints: A weighted multi-criteria approach. [preprint]
- Rai, K., Roy, A., Dattner, I., **Deb, S.** (2026). A nonparametric approach to understand multivariate quantile dynamics in financial time series. [preprint]
- Rai, K., Anand, A., **Deb, S.** (2026). Multiperiod volatility forecasting with optimization-based model selection: Evidence from NIFTY-50 Banks. [preprint]
- Gupta, K., Srivastava, S., **Deb, S.**, Panchapagesan, V. (2025). CPRI-Office: A new commercial property rental index for Indian cities using spatio-temporal modeling techniques. [preprint]
- **Deb, S**, Neves, C, Roy, S (2024) Nonparametric quantile regression for spatio-temporal processes. [preprint]
- Roy, A., Podder, M., **Deb, S.** (2024). Nonparametric method of structural break detection in stochastic time series regression model. [preprint]
- Rawat, S., Durrant, A., Simpson, A., Nielson, G., Berrett, C, **Deb, S** (2023) A Bayesian approach to identify changepoints in spatio-temporal ordered categorical data: An application to COVID-19 data. [preprint]
- Bag, S., Gupta, K., **Deb, S.** (2022). A review and recommendations on variable selection methods in regression models for binary data. [preprint]

Journal & conference publications

- Soni, A., **Deb, S.**, Das, D. (2026). Regime-Dependent Value-at-Risk and Risk Spillovers in Multivariate Cryptocurrency Markets. Applied Stochastic Models In Business And Industry, 42(6), e70140. [publication]
- **Deb, S.** (2026). Skill or chance? A Bayesian analysis of dependence and heterogeneity in penalty shootouts in football. Journal of Sports Analytics, 12. [publication]

- Panja, M., Chakraborty, T., Biswas, A., **Deb, S** (2026) E-STGCN: extreme spatio-temporal graph convolutional networks for air quality forecasting. Journal of the Royal Statistical Society Series A, qnag010. [publication]
- **Deb, S.**, Das, S. (2026). Optimal selection of the starting lineup for a football team. IIMB Management Review, 100642. [publication]
- Basu Sarbadhikary, S., Roy, A., **Deb, S.** (2026) A data-driven approach to spatial zoning and anomaly detection in the dynamic real estate network. Environment and Planning B: Urban Analytics and City Science, 23998083251411954. [publication]
- Sen, S., **Deb, S.** (2025) tSNE-Spec: A new classification method for multivariate time series data. Journal of Multivariate Analysis, 105537. [publication]
- Gupta, K., **Deb, S.** (2025) A divide-and-conquer approach for spatio-temporal analysis of large house price data from Greater London. Annals of Operations Research, 1-42. [publication]
- Deshmukh, S. S., **Deb, S.** (2025) A survey of statistical and machine learning methods of quantile regression in time series and their suitability in predicting dengue outbreaks. Japanese Journal of Statistics and Data Science, 1-49. [publication]
- **Deb, S.**, Jana, K. (2024) Nonparametric quantile regression for time series with replicated observations and its application to climate data. Statistical Science, 39(3), 428-448. [publication]
- **Deb, S.**, Roy, R., Das, S. (2024). Forecasting elections from partial information using a Bayesian model for a multinomial sequence of data. Journal of Forecasting, 43(6), 1814-1834. [publication].
- Divekar, C., **Deb, S.**, Roy, R. (2024) Real-time forecasting within soccer matches through a Bayesian lens. Journal of the Royal Statistical Society Series A: Statistics in Society, 187(2), 513–540. [publication]
- Chattopadhyay, A., **Deb, S.** (2024) A spatio-temporal model for binary data and its application in analyzing the levels of COVID-19 spread. AStA Advances in Statistical Analysis, 108(4), 823-851. [publication]
- Lakshmi M.V., **Deb, S.**, Sen, R. (2024) Environmentally Responsible Index Tracking: Maintaining Performance while Reducing Carbon Footprint of the Portfolio. Statistics and Applications (forthcoming).
- Gupta, K., Krishnamurthy, V., **Deb, S.** (2024) What elements of the opening set influence the outcome of a tennis match? An in-depth analysis of Wimbledon data. IIMB Management Review. [publication]
- Roy, A., **Deb, S.**, Chakarwarti, D. (2024) Impact of COVID-19 on public social life and mental health: A statistical study of Google Trends data from the USA. J. Appl. Stat., 51(3), 581-605. [publication]
- Roy, A., Soni, A., **Deb, S.** (2023) A wavelet-based methodology to compare the impact of pandemic versus Russia-Ukraine conflict on crude oil sector and its interconnectedness with other energy and non-energy markets. Energy Economics, 124, 106830. [publication]
- **Deb, S.**, Karmakar, S. (2023) A novel spatio-temporal clustering algorithm with applications on COVID-19 data from the United States. Computational Statistics & Data Analysis, 107810. [publication]
- **Deb, S.** (2023). Analyzing airlines stock price volatility during COVID-19 pandemic through internet search data. International Journal of Finance & Economics, 28(2), 1497-1513. [publication]
- Mareeswaran, M., Sen, S., **Deb, S.** (2023). New methods of structural break detection and an ensemble approach to analyze exchange rate volatility of Indian rupee during COVID-19. Journal of the Royal Statistical Society Series A: Statistics in Society, 187(1), 39-61. [publication]
- Rawat, S., **Deb, S.** (2023). A spatio-temporal statistical model to analyze COVID-19 spread in the USA. Journal of Applied Statistics, 50(11-12), 2310-2329. [publication]
- **Deb, S.**, Majumdar, M. (2023). A quadratic trend-based time series method to analyze the early incidence pattern of COVID-19. Biostatistics & Epidemiology, 7(1), e2076529. [publication]
- Majumdar, M., Banerjee, M., Sengupta, J., **Deb, S.**, Jana, C. K., Roy, B. K. (2023) Prevalence and spectrum of diabetic peripheral neuropathy and its correlation with insulin resistance – An experience from eastern India. International Journal of Advanced Research, 11(06), 1085-1094. [publication]
- **Deb, S.**, Deb, S. (2022). An ensemble method for early prediction of dengue outbreak. Journal of the Royal Statistical Society Series A, 185(1), 84-101. [publication]
- **Deb, S.** (2022). A goal based index to analyze the competitive balance of a football league. Journal of Quantitative Analysis in Sports, 18(3), 171-186. [publication]

- Nahata, S., **Deb, S.** (2021) A machine learning approach to analyze the effect of situational and player-dependent features on converting freekicks in soccer. Asia-Singapore Conference on Sport Science. [publication]
- **Deb, S.**, Tsay, R. S. (2019). Spatio-temporal Models with Space-time Interaction and Their Applications to Air Pollution Data. Statistica Sinica, 29, 1181-1207. [publication]
- **Deb, S.**, Dey, D. (2019). Spatial Modeling of Shot Conversion in Soccer to Single out Goalscoring Ability. Journal of Sports Analytics, 5(4), 281-297. [publication]
- **Deb, S.** (2019). VAR Model Based Clustering Method for Multivariate Time Series Data. Journal of Mathematical Sciences, 237(6), 754-765. [publication]
- Prickett, K.C., Guiterrez, C., **Deb, S.** (2019). Family Firearm Ownership and Firearm-related Mortality among Young Children: 1976-2016. Pediatrics, 143(2), e20181171. [publication]
- Chazin, H., **Deb, S.**, Falk, J., Srinivasan, A. (2019). New Statistical Approaches to Intra-individual Isotopic Analysis and Modelling of Birth Seasonality in Studies of Herd Animals. Archaeometry, 61(2), 478-493. [publication]
- **Deb, S.**, Pourahmadi, M., Wu, W. B. (2017). An Asymptotic Theory for Spectral Analysis of Random Fields. Electronic Journal of Statistics, Vol. 11, No. 2, p. 4297-4322. [publication]
- Zechner, C., **Deb, S**., Koeppl, H. (2013). Marginal Dynamics of Stochastic Biochemical Networks in Random Environments. In Control Conference (ECC), 2013 European, p. 4269-4274, IEEE. [publication]

Teaching

Degree-granting Programmes, at Indian Institute of Management Bangalore:

- Multivariate Statistics (Core for PhD students) — Spring, 2021 – 2026
- Nonparametric Statistics (Core for PhD students) — Spring, 2025
- Decision Sciences I (Core for MBA students) — Fall, 2020 – 2026
- Time Series & Forecasting Techniques (Core for MBA students) — Spring, 2026
- Sports Analytics (Elective for MBA students) — Fall, 2023 – 2026
- Probability and Statistics 2 (Core for BS students) — Spring, 2026
- Advanced Statistics for Business (Core for BBA students) — Winter, 2025
- Advanced Statistical Methods & Computing (Core for PhD students) — Fall, 2024
- Statistical Inference (Core for PhD students) — Winter, 2020

Executive Education Programmes, at Indian Institute of Management Bangalore:

- Data Science & Artificial Intelligence (Co-director) — 2025 – 2027
- IIMB-CREDAI Business Leadership Programme (Co-director) — 2025 – 2027
- MDP for Food Corporation of India (8 sessions) — Apr 2024
- Business Analytics and Intelligence (3 sessions) — Fall, 2022 – 2024

Other courses:

- Advanced Statistical Analysis (EEP), Indian School of Business — 2026
- Introductory Statistics (UG), University of Chicago — 2015, 2017
- Statistical Models and Methods I (UG), University of Chicago — 2015

Consulting services

National Foundation for Corporate Governance, Ministry of Corporate Affairs, India 2025 – 2027

- Topic: Feedback survey and concurrent evaluation of the Prime Minister Internship Scheme.

Smart City Mission, Ministry of Housing & Urban Affairs, India 2024

- Topic: Improvement in quality of education through smart classrooms.

DBS Bank, Singapore 2023

- Topic: Value of data joint study.

SportsKPI, Bengaluru, India 2022-23

- Topic: Valuation of a Kabaddi player and its use in building strategies.

Latest seminars

- **June 2026,** Nonparametric regression of spatio-temporal data using infinite-dimensional covariates. IMS Asia Pacific Rim Meeting 2026 (IMS-APRM), Hong Kong.

- **May 2026**. Optimizing Football Squad Selection: A Multi-Objective Approach to Transfer Strategy under Budget Constraints. AUEB Sports Analytics Workshop & Conference 2026 (SAW 2026), Athens, Greece.
- **Dec 2025**. Bridging search behavior and market dynamics: A hybrid model for high-frequency financial data. CFE-CMStatistics Conference, London, UK.
- **Dec 2025**. Real-time forecasting in football matches. Liverpool Management School, Liverpool, UK.
- **May 2025**. Data Handling, Visualisation, and Storytelling in Academic Research. Workshop-cum-training session at RV University, Bengaluru, India.
- **Dec 2024**. Nonparametric Regression of Spatio-temporal Data using Infinite-dimensional Covariates. IISA Conference 2024, Cochin, India.
- **Dec 2024**. Structural breaks in the spatial network of real estate dynamics: A study of UK property transactions. Essex Data Science Seminar Series, University of Essex, Colchester, UK.
- **Sep 2024**. Nonparametric changepoint detection in time series data. COMPSTAT 2024, Giessen, Germany.
- **Jul 2024**. A workshop on 'Data summarisation and Visual Storytelling using Statistics', PES University Research Workshop Series, Bengaluru, India.
- **Jun 2024**. Real-time forecasting in sports. AI Frontiers Series, Commonwealth Bank of Australia (virtual).
- **May 2024**. A workshop on Sports Analytics. Conference on Excellence in Research & Education, Indian Institute of Management Indore, India.

Honors

At Indian Institute of Management Bangalore:

- Excellence in Teaching Recognition — Multiple years since 2022

At University of Chicago:

- International House Ralph W. Nicholas Fellowship — 2017-18
- Graduate Council Travel Fund Award — 2017
- Runner-up for Department of Statistics Consulting Award — 2016
- Nominated for Best Teaching Assistant in Physical Sciences Division — Winter 2014

Earlier awards:

- Kishore Vaigyanik Protsahan Yojana scholarship, Indian Institute of Science — 2007 to 2013
- Selected for International Mathematical Olympiad Training Camp, India — 2007 & 2008

Professional Services

Administrative roles, at Indian Institute of Management Bangalore:

- Chairperson, Decision Sciences Area. — Apr 2024 - present
- Member, Postgraduate Programme Committee. — Apr 2025 - present
- Doctoral Program Coordinator, Decision Sciences Area. — 2023-24
- Mentor in Pre-doctoral Programme, Decision Sciences Area. — 2021 - 2024

Editorial Board member, for the following journals:

- BMC Public Health, Associate Editor. — 2026 – present
- Journal of Sports Analytics, Associate Editor. — 2025 – present
- Sankhya B, Associate Editor. — 2025 – present
- Nature Scientific Reports, Associate Editor in the Climate Change track. — 2024 – present

Reviewer, for many journals including the following:

- Annals of Applied Statistics, Biometrics, Electronic Journal of Statistics, Journal of Multivariate Analysis
- Journal of the American Statistical Association, Journal of the Royal Statistical Society
- Journal of Sports Analytics, Sankhya A and Sankhya B, Statistics and Probability Letters

Skills

Technical strength:

- Proficient: R, Python, LaTeX, Microsoft Office, Google Suite
- Working knowledge: MATLAB, SQL, C, C++

Languages:

- Bilingual: English, Bengali; Intermediate: Hindi; Basic: Portuguese.